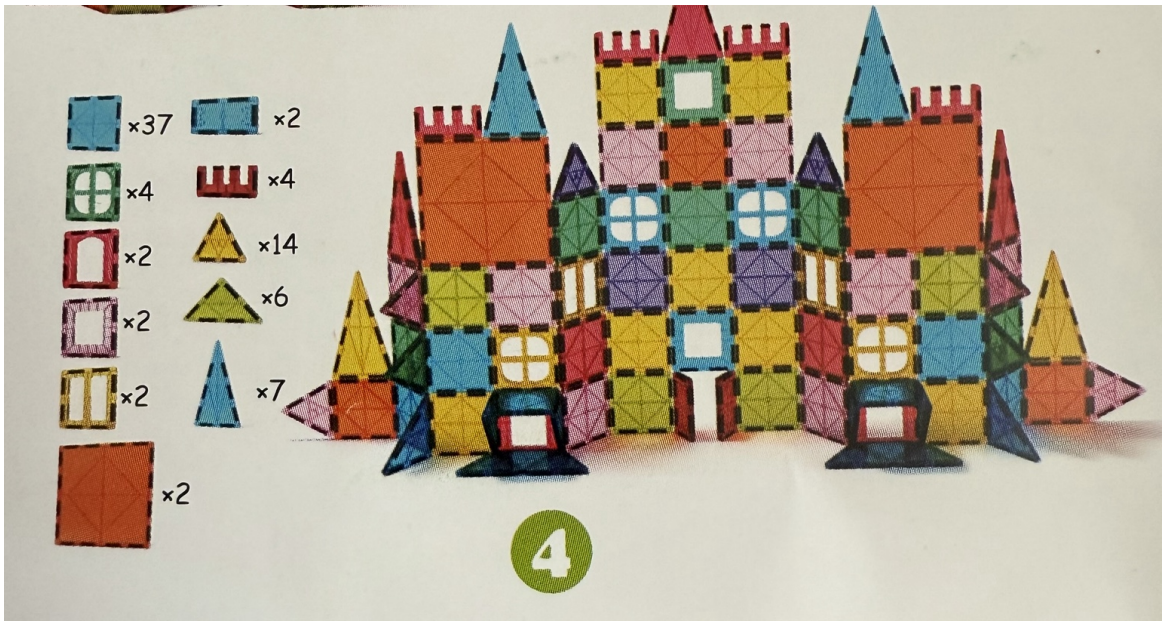
Toy Castle Image Preservation Demo: Same Object vs. Plausible Replacement

Short Visual Audit Report | Prepared as a Grounded DI LLC visual-fidelity demo

Purpose. Document a simple image-generation preservation test using a children's magnetic-tile castle instruction image. The test was run and framed as a Grounded DI visual-audit exercise: not whether a model can make a pretty castle, but whether it preserves a constrained source image when asked to keep the toy the same.

> **Exact test prompt submitted:**
> **"Keep the toy the same but render it so it's easier to see"**

Core issue. The source image is not merely a picture of a toy. It is a visual specification. It contains the completed magnetic-tile castle, a parts/count legend on the left, the green "4" instruction badge, specific tile arrangements and colors, and checkable inventory constraints. Because the image includes the answer key, a faithful output should preserve both the build and the parts/count metadata.

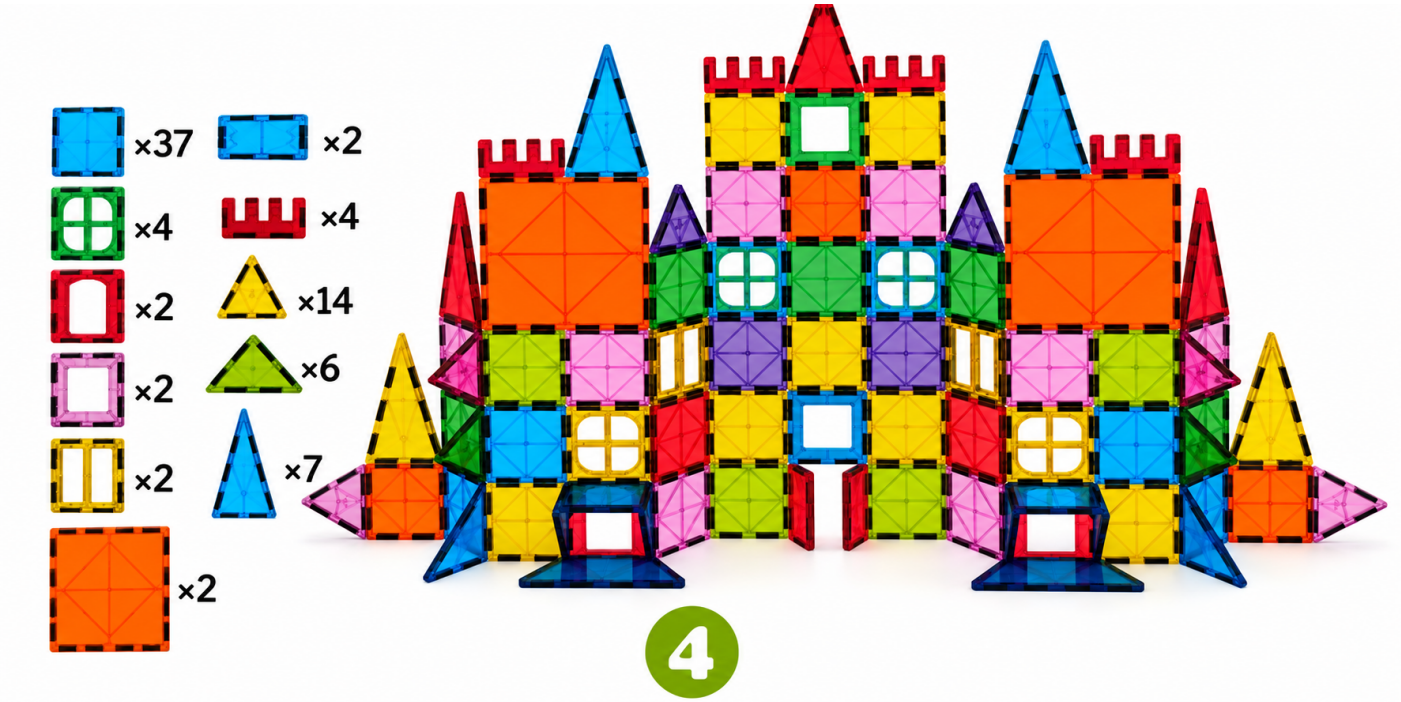


Source instruction image submitted for the preservation test.

What must be preserved: the full castle build, the parts/count legend, the green "4" badge, and the specific tile geometry: windows, towers, triangles, side wings, colors, and symmetry.

A. Grounded DI Implementation Output

Grounded DI's run produced a high-visibility full-page rendering that retained the instruction-page concept, including the left-side parts legend and the green "4" badge. Visually, this is the strongest full-page preservation result in the set. It still should not be treated as construction-accurate without a piece-by-piece audit against the original inventory.



Grounded DI / implementation output: strong full-page clarity, but construction fidelity remains audit-dependent.

Observed result: strong visual result / audit needed. It preserves more context than the other outputs, but it is still a generated render, not verified assembly instructions.

B. Grok Output

Grok's process visibly searched for magnetic-tile castle references and generated a clean but different castle. The output is visually appealing, but it replaced the source build with a generic magnetic-tile castle and removed the instruction metadata.

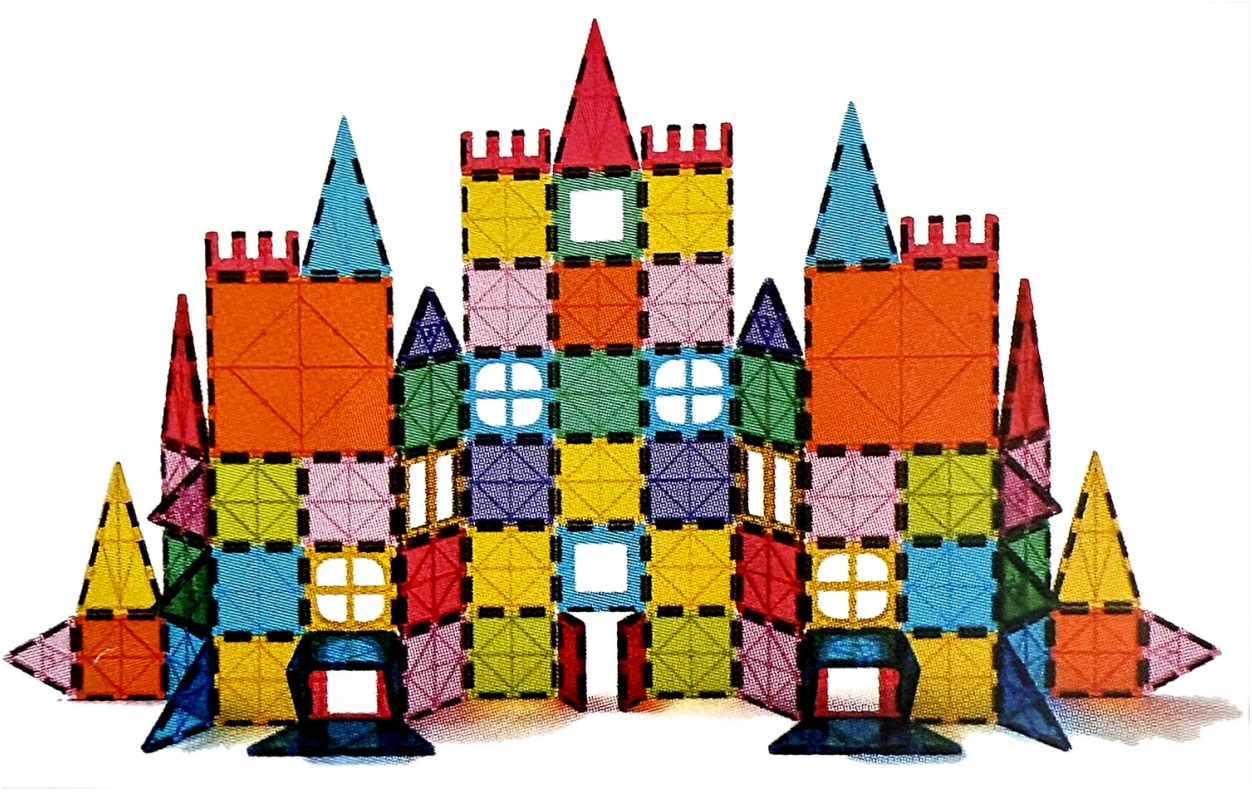


Grok submission/output screenshot: the generated castle is cleaner but not the same build.

Observed result: generic castle replacement. The object category is preserved; the actual visual specification is not.

C. Claude Output

Claude produced a cleaned-up crop/upscale of the original castle. This is closer to the source geometry than a generic re-render, but it removed the parts/count legend and the green “4” badge. It is best understood as a forensic cleanup of the castle portion, not a full preservation render of the instruction image.

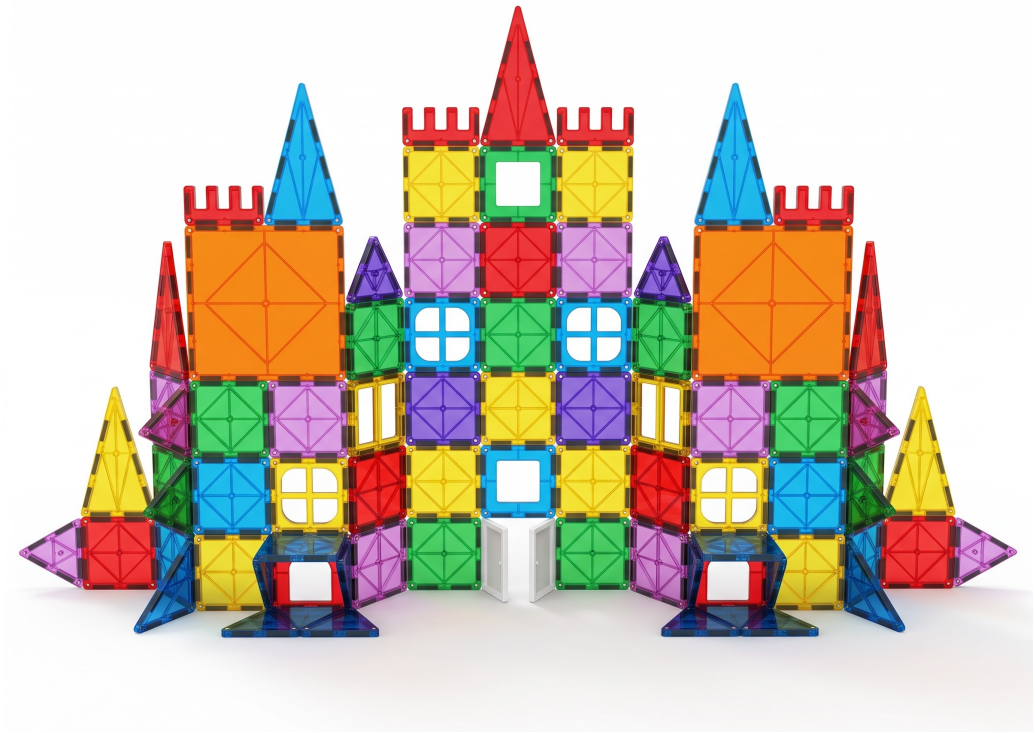


Claude output: close to the source castle, but cropped away instruction metadata.

Observed result: partial pass. Closest to the source castle form, but not a true full-image render and not metadata-preserving.

D. Gemini Output

Gemini produced a clean product-style render and later, when directly asked “Did you cheat?”, described using image-to-image shortcuts/stenciling and acknowledged geometry simplification. Visually, the output is closer than a generic replacement, but it still normalizes the geometry and alters the build.



Gemini output: clean render, but geometry and inventory constraints are simplified.

Fairness note. Gemini was the only system directly asked a follow-up self-audit question (“Did you cheat?”). The other systems were not asked that same follow-up in the same way. Gemini’s admission should therefore be treated as a product of the additional prompt, not as proof that Gemini was uniquely less faithful.

What Happened

Grok: searched/reference-blended and produced a plausible magnetic-tile castle rather than the same build.

Claude: produced a cleaned-up crop/upscale closer to the source, but removed the legend and the green “4” badge.

Gemini: produced a clean render and, after a direct follow-up, admitted using shortcuts/stenciling and simplifying geometry.

Grounded DI: produced the strongest full-page visual render, preserving more of the instruction-sheet context, but still requiring a piece-by-piece audit before being called construction-accurate.

Key Finding

The prompt was not unfair. A regular person would reasonably expect “keep the toy the same” to mean preserve the actual build and improve visibility only. The hard part is that the source image is a specification: the parts/count legend is an answer key, not decoration.

> **Outputs that look cleaner but do not preserve or reconcile the legend are illustrative replacements, not faithful renderings.**

Failure Modes Observed

• semantic replacement: making a plausible castle instead of the same castle

• geometry normalization and invented symmetry

• tile fusion / mega-piece creation

• removal of instruction metadata

• false confidence in preservation

• web/reference blending

• failure to distinguish “cleaner image” from “new image”

• failure to honor the parts/count legend as an answer key

Why This Matters

Children’s toys, mazes, worksheets, and build instructions are surprisingly strong tests of visual fidelity because the small details are not decoration. They are the task. A model either preserves the constrained visual record, or it substitutes a plausible-looking object.

Bottom-Line Verdict

System	Result	Preservation Grade	Notes
Grok	Generic castle replacement	Fail	Visually nice, wrong object
Claude	Forensic crop/cleanup	Partial pass	Closest to source castle, not a true full render, removed context
Gemini	Clean render with admitted shortcuts	Partial fail	Good self-audit after direct follow-up; geometry changed
Grounded DI	Best full-page render	Strong visual result / audit needed	Preserved more context, still not proven build-accurate

Conclusion

This demo shows that “make it clearer but keep it the same” is a hard preservation task. The issue is not whether the output looks good; the issue is whether it preserves the source object under constraint. Here, the original instruction image supplied its own answer key through the parts/count legend, so preservation should be evaluated against that specification, not against whether the result looks like a plausible toy castle.

Grounded DI’s role was to run, frame, and implement the visual audit. The report therefore uses Grounded DI as the baseline implementation label rather than a generic ChatGPT/OpenAI label.